



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2025

CS1004-1 – INTRODUCTION TO C PROGRAMMING

.(For AC, EC, EE, VL)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What should be the output:

```
int main()
{
    int a = 10.5;
    printf("%d",a);
    return 0;
}
```

A) 10.5
B) 10
C) 0
D) Compilation error
- An operator that acts on only one kind of operand is known as
A) Binary Operator
B) Unary operator
C) Ternary Operator
D) Special Operator
- Which of the following is not an arithmetic operation?
A) $a * = 10;$
B) $a / = 10;$
C) $a ! = 10;$
D) $a \% = 10;$
- What will be the output of the following C code?

```
#include <stdio.h>
int main()
{
    int a = 1, b = 1, c;
    c = a++ + b;
    printf("%d, %d", a, b);
}
```

A) a = 1, b = 1
B) a = 2, b = 1
C) a = 1, b = 2
D) a = 2, b = 2
- Which of the following is not a conditional branching statement of C language?
A) if statement
B) if...else statement
C) switch statement
D) goto statement
- In which scenario would you use the continue statement?
A) To exit from a loop early and execute the code after the loop.
B) To skip the remaining statements in the current iteration of a loop and proceed to the next iteration.
C) To jump to a specific line of code unconditionally.
D) To break out of a loop or switch statement.
- Which of the following statements is true?
A) for loop is faster than while loop
B) while loop is faster than for loop
C) do-while loop is faster than both for and while loops
D) There is no difference in execution speed between for, while, and do-while loops

8. Which one is true about the break statement in a switch case statement?
 A) To skip the current case B) To end the program
 C) To exit the switch case statement D) To exit the loop
9. To access the third element of an array named 'arr', you would use:
 A) arr[2] B) arr[3]
 C) arr(2) D) arr(3)
10. Which of the following correctly initializes an array of size 5 with all values set to 0?
 A) int arr[5] = {0}; B) int arr[5];
 C) int arr[5] = {}; D) int arr[] = {0,0,0,0,0};
11. In C, what is a function primarily used for?
 A) Decision making B) Variable declaration
 C) Code organization and reusability D) Printing output
12. Which of the following is a correct way to define a function in C?
 A) int sum(int a, int b) { return a + b; } B) function sum(int a, int b) { return a + b; }
 C) sum(int a, int b): int { return a + b; } D) int sum(a, b) { return a + b; }
13. Which function changes all the characters of a string to lowercase?
 A)strupr() B)strrev()
 C)strlwr() D)strcpy()
14. What does putchar() function do?
 A) Reads a character from input B) Prints a string to the output
 C) Prints a single character to the output D) Reads a string from input
15. What is the difference between strcat() and strncat()?
 A) strcat() copies characters, while strncat() concatenates characters B) strcat() appends a string, while strncat() appends a fixed number of characters
 C) strcat() compares strings, while strncat() copies strings D) There is no difference; they are identical
16. What does the strcmp() function return when two strings are equal?
 A) 0 B) 1
 C) strlen() D) strrev()
17. How is the size of a structure determined in C?
 A) By the size of its largest member B) By the total size of its members
 C) By the size of its smallest member D) By the number of members
18. What does the fclose() function return if the file is successfully closed?
 A) 0 B) 1
 C) EOF D) -1
19. How do you read a character from a file in C?
 A) readchar() B) getc()
 C) read() D) getchar()
20. How do you declare an array of structures in C for a structure named myStruct?
 A) array myStruct arr[]; B) myStruct arr;
 C) struct array myStruct[]; D) myStruct array[];

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Discuss the following data types, including their general syntax, memory usage, and examples:
- | | | | | |
|-----------|---|-----|---|---|
| i. int | 6 | L*2 | 1 | 1 |
| ii. float | 6 | L3 | 1 | 1 |
| iii. char | 4 | L2 | 2 | 1 |
- b) Build a C program to read 3 numbers and compute the total, average of these numbers.
- c) Describe the if-else statement, providing its syntax and a corresponding flowchart to illustrate its execution flow.

- | | | | | | |
|-------|--|---|----|---|---|
| 2. a) | Explain the following operators with examples | | | | |
| | i. Logical AND and Logical OR operator | | | | |
| | ii. Bitwise left shift and Bitwise right shift operator | 6 | L2 | 1 | 1 |
| b) | Explain the else-if ladder, including its syntax, flowchart, and an example. | 6 | L2 | 2 | 1 |
| c) | Build a C program to compute the sum of the individual digits of a given number using a while loop. | 4 | L3 | 2 | 1 |
| 3. a) | Explain the importance of operator precedence and associativity in evaluating arithmetic expressions. Evaluate the following expressions step by step: | | | | |
| | i. $(10 + 5) * 2 - 20 / (4 \% 3)$ | | | | |
| | ii. $(20 \% 7) * 3 + 4 * 2 - 10 / 5$ | 6 | L2 | 1 | 1 |
| b) | Build a C program to compute the GCD (Greatest Common Divisor) and LCM (Least Common Multiple) of two given numbers using Euclid's method. | 6 | L3 | 2 | 1 |
| c) | Distinguish between entry-controlled loops and exit-controlled loops with explanations. | 4 | L2 | 2 | 1 |

Unit – II

- | | | | | | |
|-------|---|---|----|---|---|
| 4. a) | Explain the declaration and initialization of a one-dimensional array in C with proper syntax and an example. | 6 | L2 | 3 | 1 |
| b) | Build a C program to search a given key element using Binary search technique. | 6 | L3 | 3 | 1 |
| c) | Develop a C program to perform the following operations using string built-in functions: | | | | |
| | i. reverse a given string | | | | |
| | ii. to find the length of the given string. | 4 | L2 | 4 | 1 |
| 5. a) | Discuss the following string manipulation functions with their syntax and examples: | | | | |
| | i. strcmp() | | | | |
| | ii. strcpy() | | | | |
| | iii. strncpy() | 6 | L2 | 4 | 1 |
| b) | Build a C program, to implement a bubble sort technique using function to sort given N integers. | 6 | L3 | 3 | 1 |
| c) | Explain how strings are declared and initialized in C with an example. | 4 | L2 | 4 | 1 |
| 6. a) | Discuss the following categories of functions with syntax and examples: | | | | |
| | i. Function with argument and return type | | | | |
| | ii. Function without argument and no return type | 6 | L2 | 3 | 1 |
| b) | Build a C program to transpose a matrix of order M x N and find the trace of the resultant matrix. | 6 | L3 | 3 | 1 |
| c) | Demonstrate the usage of the strcat() function with an example program. | 4 | L2 | 4 | 1 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|---|
| 7. a) | Define a Structure in C and explain how to declare, define, and access structure members with an example. | 6 | L2 | 5 | 1 |
| b) | Develop a C program that reads user input from the console, stores the input in a file, and then reads and displays the stored content on the terminal. | 6 | L3 | 5 | 1 |
| c) | Define a pointer in C and explain the significance of the reference (&) operator and dereference (*) operator with examples. | 4 | L2 | 5 | 1 |

8. a) Explain the following file operations with general syntax and an example:
- i. Opening and closing a file
 - ii. Reading a file
 - iii. Writing to a file
- b) Develop a C program that declares a structure called Patient to hold a patient's information, including their name, age, weight (in kilograms), and height (in meters). The program should request these details from the user and compute the Body Mass Index (BMI) using the formula $BMI = \text{weight} / (\text{height}^2)$. After performing the calculation, it should display the patient's details along with the computed BMI.
- c) Build a C program to input and print array elements and compile sum of elements using pointers.

6	L2	5	1
6	L3	5	1
4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
